

Latent Bridge Matching: Efficient OT-Coupled Flow Matching for Domain-Level Artistic Transfer

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Abstract

Domain-level artistic transfer must balance three objectives that often conflict: target-style strength, content preservation, and computational cost. We propose Latent Bridge Matching (LBM), a compact latent-space framework that treats stylization as controlled transport in a VAE latent manifold rather than as direct latent reconstruction. The model couples mini-batch optimal-transport endpoint assignment, time-conditioned transport prediction, semantic routing, and Semantic-Aligned Sliced Wasserstein (SA-SWD) terminal alignment. In this view, the style tokenizer specifies what target style should be executed, while the LANCET backbone realizes that control as a latent vector field. Under a strict 750-output protocol over five domains, a 3.9M-parameter LBM model reaches near-SaMST CLIP-style (0.716 vs. 0.719), improves LPIPS and EC, and shows a favorable measured operating-point wall-clock profile in our reproduced records: 310s to the reported checkpoint versus 6769s for the reproduced SaMST operating point, with strict-750 inference in 85.4s (114ms/image) on a single RTX 4070 Laptop GPU. On a Distinct5-512 stress split built from the five most separated WikiArt style domains under a CLIP-style screening criterion and used only as a metric stress test, the best LBM checkpoints define the strongest measured content-preserving frontier among the evaluated methods: LBM-F reaches CLIP-S 0.6969 and LPIPS 0.3186 after about 1.2 minutes of training, while SaMAM-2250 reaches 0.5811 and 0.3538 after about 7.6 hours, and SaMST-512 attains higher raw style only with severe LPIPS and ArtFID degradation. Taken together, the results show that raw CLIP-style is sensitive to the unchanged-image prior on separated art-to-art transfer, and that the main remaining bottleneck is not tokenizer size, but faithful style execution through the latent renderer.

Introduction

Artistic style transfer has long served as a testbed for separating image content from visual appearance. Classical neural style transfer showed that convolutional features and Gram statistics can synthesize compelling artistic textures (Gatys, Ecker, and Bethge 2016), while feed-forward arbitrary style transfer made real-time deployment practical through normalization-based style

modulation (Huang and Belongie 2017). Subsequent attention and transformer systems improved local correspondence and long-range context (Park and Lee 2019; Deng et al. 2022; Zheng et al. 2024). Recent reference-guided stylization then diversified along three axes: perceptual-quality refinement in AesPA/AesFA (Hong et al. 2023; Huang et al. 2024), more structured style injection in HSI (Zhang et al. 2025), and semantic-region consistency in SCSA (Shang et al. 2025). In parallel, training-free VQ and diffusion-based methods strengthened style expression without retraining (Gim, Yoo, and Uh 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024; Jiang et al. 2025). However, practical multi-style artistic transfer still exposes a persistent three-way tension: strong target-style expression, faithful content preservation, and low computational cost.

This tension is especially visible in domain-level multi-style transfer. Many recent methods are still reference-image-guided: they receive a style exemplar at inference and focus on better content-style correspondence or stronger style injection (Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025; Chung, Hyun, and Heo 2024; Deng et al. 2024). Compact multi-style methods instead amortize a fixed style family into pluggable codes or lightweight branches (Liu et al. 2024), while recent state-space models seek similar efficiency improvements for reference-guided transfer through alternative global-mixing operators (Liu et al. 2025; Botti et al. 2025). Our setting is closer to the compact multi-style regime: inference receives only a target style/domain identifier, not a target style image. In this regime, the bottleneck is not exemplar matching but executing a stable style control signal without washing out content or collapsing into texture artifacts. Under our strict 750-output protocol, StyleID attains the highest raw CLIP-style score but exhibits severe content collapse, while SaMST achieves strong raw style and content scores yet produces visible muddy and grain-like texture artifacts. These observations suggest that standard style/content metrics alone are insufficient. A competitive system must also be evaluated for perceptual quality and artifact-sensitive texture realism. In particular, low-frequency structure metrics can overestimate perceptual quality when local texture becomes muddy or

grain-like.

A deeper source of this tension is geometric. VAE latents are compressed feature coordinates with non-trivial local curvature, not RGB pixels. In this regime, pointwise Euclidean latent endpoint reconstruction encourages mean solutions, suppresses semantically valid local rearrangements, and blurs the distinction between “preserving content” and “doing nothing.” Our current claim is intentionally narrow: the evidence in this paper supports replacing latent endpoint reconstruction with OT-coupled endpoint construction and Wasserstein-style terminal matching, while keeping path regularity explicit through velocity-field dynamics. Broader claims about the optimal local residual geometry of every transport loss term are left open.

We propose Latent Bridge Matching (LBM), which models stylization as controlled latent transport rather than direct texture synthesis. Instead of directly predicting RGB images, the method operates on compact VAE latents with three integrated components: (i) a mini-batch entropic OT coupling that matches unpaired content and target-style latents via sliced-Wasserstein cost; (ii) a time-conditioned latent vector field trained to predict transport updates under endpoint-coupled bridge supervision; and (iii) SA-SWD, a semantic-aligned terminal regularizer that pulls the Euler-integrated endpoint toward target-style patch statistics using projection axes selected from the semantic cross-attention keys. A learnable style spatial prior and semantic cross-attention provide efficient domain-level conditioning without per-image optimization or diffusion sampling.

Our contributions are:

- A structured latent-transport formulation for domain-level artistic transfer, decomposing unpaired supervision into OT-based endpoint construction, transport-field learning, and semantic-aligned terminal endpoint regularization.
- A design-grounding formal analysis connecting the training objective to inference-time behavior: a path-action decomposition theorem, an $O(1/K)$ Euler discretization bound, and a Lipschitz-continuity result for OT-based endpoint assignment—each paired with direct empirical validation.
- Semantic-Aligned Sliced Wasserstein (SA-SWD), which converts cross-attention routing evidence into terminal distribution-matching axes, avoiding purely isotropic patch matching while retaining the efficiency of SWD.
- An executable style-tokenizer formulation and diagnostic protocol, separating the style-side representation from the latent renderer. The tokenizer receives style identity and learned style statistics, never per-sample target evidence in the main style-id protocol; representation probes then test whether learned controls remain distinct after the LANCET vector field executes them.

- A measured operating-point efficiency-quality result against the closest efficient baseline SaMST: near-parity CLIP-style (0.716 vs. 0.719), improved LPIPS (0.451 vs. 0.466), higher EC (0.393 vs. 0.384), a shorter reproduced wall-clock time to the reported operating point (310 s vs. 6769 s), and a practical inference profile of 85.4 s per strict-750 evaluation pass.
- A separated Distinct5-512 stress benchmark, used as a metric stress test rather than a universal art benchmark, on which the best LBM variants define the strongest measured content-preserving frontier among the evaluated methods.
- Consistent gains over the heavier S2WAT baseline: higher CLIP-style, substantially lower LPIPS, higher EC, and a $16.7\times$ smaller model—showing that the efficiency gain is not obtained by sacrificing transfer quality.

Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses (Gatys, Ecker, and Bethge 2016). AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics (Huang and Belongie 2017). Attention-based methods such as SANet and StyTr2 improved local style matching and long-range interaction through learned content-style correspondence (Park and Lee 2019; Deng et al. 2022). More recent reference-guided arbitrary stylization continues to refine the injection path itself: PuffNet emphasizes cleaner content/style feature separation, HSI replaces point-wise attention with a holistic injector, and SCSA focuses on semantic-region consistency in arbitrary semantic style transfer (Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025). These works mainly ask how a style exemplar should be fused at inference time. Our problem is different: the main protocol in this paper does not provide a target style image at inference, so the central question is how a domain-level style control should be executed without exemplar-conditioned correspondence.

Efficient multi-style transfer and compact style representations. Multi-style transfer amortizes many styles into one compact model rather than solving a fresh arbitrary-style problem for every target image. SaMST is the closest recent representative: it separates style modeling from style transfer through pluggable style representations and a style-aware transfer network, explicitly targeting the quality-efficiency trade-off for many-style deployment (Liu et al. 2024). This line is the most relevant tokenizer-side baseline for our work. Our contribution is not another pluggable codebook alone. Instead, we ask whether a compact style representation remains executable after it passes through a latent transport field, and we diagnose the resulting bottleneck as renderer-side style execution rather than raw tokenizer size.

State-space and Mamba-based style transfer. Recent work also revisits style transfer through state-space backbones. Mamba-ST and SaMam argue that linear-complexity state-space operators can recover global receptive fields more efficiently than quadratic attention, improving the efficiency-quality frontier of arbitrary or many-style transfer (Botti et al. 2025; Liu et al. 2025). These papers are important because they shift the architectural question from “CNN or transformer” to “which global-mixing primitive is most efficient.” Our comparisons to SaMam take this line seriously. The distinction is that LBM does not claim a new global-mixing backbone; its main contribution is objective-side: OT-coupled endpoint construction, transport-field learning, and semantic-aligned terminal distribution matching.

Training-free and diffusion methods. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free stylization (Gim, Yoo, and Uh 2024). StyleID and Z* show that large diffusion priors can be adapted for artistic style transfer without retraining by manipulating attention features or attention weights at inference time (Chung, Hyun, and Heo 2024; Deng et al. 2024). SMS further pushes this direction by casting stylization as diffusion-side style distribution matching with additional spectrum-aware and semantic-aware refinement (Jiang et al. 2025). These methods can be visually strong, but they generally assume a style reference image and often inherit the inference cost or calibration issues of large generative priors. Our work instead targets compact retrainable latent integration with style-id conditioning rather than iterative diffusion sampling, prompt engineering, or per-style inference-time optimization.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems (Cuturi 2013; Peyré and Cuturi 2019), while the Schrodinger problem connects entropic transport to stochastic path measures (Léonard 2014). We use SWD as a lightweight terminal distribution matching objective and specialize it with semantic routing in SA-SWD. OT serves as a mini-batch endpoint-construction mechanism in latent space, while the formal analysis below characterizes the bounded behavior of the resulting transport objective rather than asserting exact global transport optimality. Evaluation itself remains unsettled in arbitrary style transfer: prior studies report inconsistent protocols across papers and show that commonly used automatic metrics are style-dependent and require calibration to human preference (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our paper builds on that critique but makes a narrower claim. We do not propose a universal replacement metric. Instead, for

separated art-to-art transfer we make the unchanged-image prior explicit through an ‘idt’ reference, then interpret CLIP-derived scores as incremental style movement rather than standalone proof of stylization. We therefore combine perceptual and distributional measures including LPIPS (Zhang et al. 2018), FID (Heusel et al. 2017), KID (Bińkowski et al. 2018), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), MANIQA (Yang et al. 2022), and ArtFID (Wright and Ommer 2022). This broader metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts or may partially reflect art-domain priors already present before stylization.

Method

Overview

LBM is best understood as a structured latent transport design for domain-level artistic transfer. Let $z_0 \in \mathbb{R}^{4 \times 32 \times 32}$ denote the VAE latent of a content image and let s denote a target style domain identifier. Across the codebase and reproduced checkpoints, the shared design has four coupled components: (i) an optimal-transport coupling that anchors unpaired content and target-style endpoints within each minibatch; (ii) a time-conditioned transport field $v_\theta(z_t, t, s)$ that predicts latent updates; (iii) kinetic regularization that discourages unnecessary latent displacement; and (iv) SA-SWD, which acts as a terminal corrective force pulling the integrated endpoint toward target patch statistics along semantically meaningful axes. At inference, the learned vector field is integrated with explicit Euler steps and decoded by the VAE decoder. In this decomposition, the tokenizer is a semantic beacon that specifies the target region of style space, and the backbone is the executor that realizes the corresponding transport field.

The paper reports two closely related training lineages. The historical bridge formulation uses random-time stochastic-bridge supervision for velocity prediction. The current Distinct5 tokenizer family, however, resolves through the codebase config loader to the OMF path (`objective_mode=omf`), whose active loss is better read as OMF-style endpoint-delta training with terminal matching and kinetic regularization. To avoid overstating closure, we make mechanism claims at the shared transport-design level unless a specific objective family is named explicitly.

Unpaired latent-domain sampling and OT coupling

We train directly on precomputed VAE latents without paired supervision. Each minibatch samples content latents $\{z_c\}$ from all source domains and target-style latents $\{z_s\}$ from the requested target domain. Identity pairs are naturally included through uniform target-domain sampling, which stabilizes the learned transport around same-domain mappings.

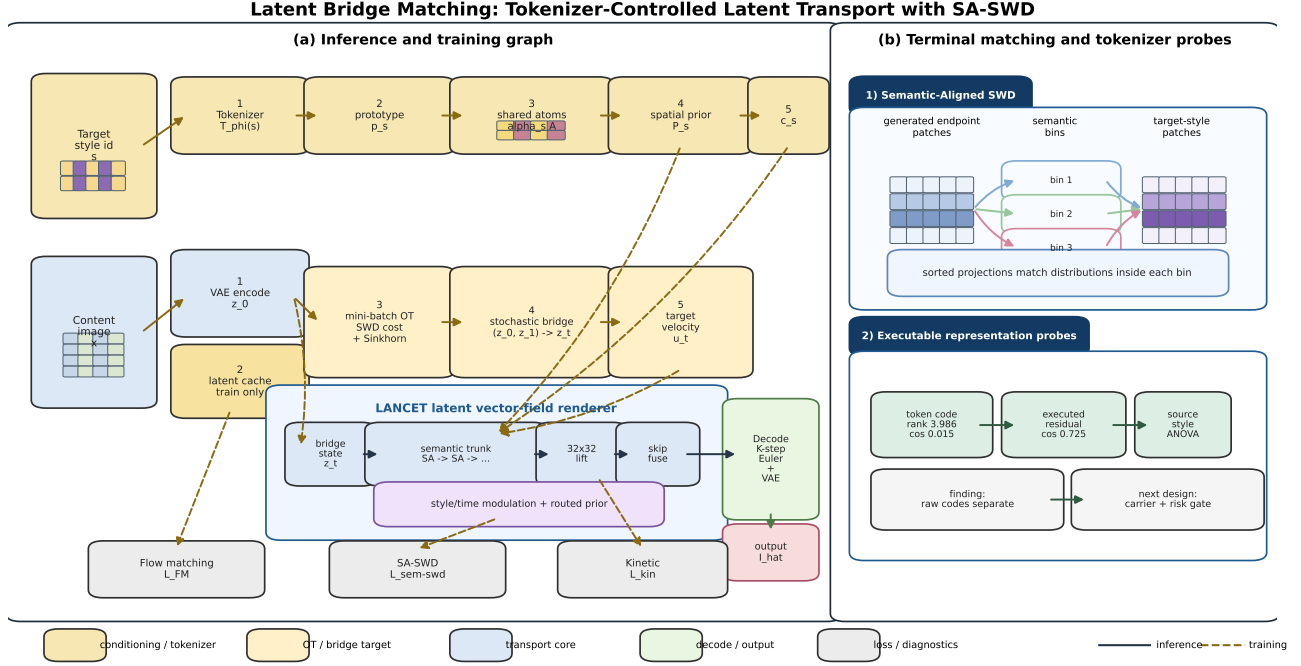


Figure 1: Overview of Latent Bridge Matching. The style tokenizer produces style-side controls from the target style identifier, and the LANCET vector field executes those controls as latent transport. Target-domain latents enter only through training-side OT endpoint construction and SA-SWD terminal matching; they are not inference-time conditioning inputs. Solid arrows denote the active inference path and dashed arrows denote supervision.

Rather than using random target assignments, we construct a mini-batch transport coupling using the sliced-Wasserstein cost. For a batch of content latents $\{z_c^{(i)}\}$ and target-style latents $\{z_s^{(j)}\}$, the SWD cost matrix is

$$C_{ij} = \text{SWD}(z_c^{(i)}, z_s^{(j)}). \quad (1)$$

An entropic optimal-transport plan is then obtained via the Sinkhorn algorithm:

$$\pi^* = \arg \min_{\pi} \sum_{ij} C_{ij} \pi_{ij} + \epsilon H(\pi), \quad (2)$$

where $H(\pi)$ is the entropy regularizer. Each content latent is matched to its coupled target $\tilde{z}_1 \sim \pi^*(z_s|z_c)$. This coupling provides style-domain endpoints without requiring spatially aligned targets.

Time-conditioned latent transport supervision

The core of the method is a time-conditioned velocity field $v_{\theta}(z_t, t, s)$ that transports latents along a stochastic bridge. At a randomly sampled time $t \sim \mathcal{U}(0, 1)$, the bridge state is

$$z_t = (1 - t)z_0 + t\tilde{z}_1 + \sigma\sqrt{t(1 - t)}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

and the target velocity is the derivative of the transport:

$$u_t = \tilde{z}_1 - z_0 + \sigma \frac{1 - 2t}{2\sqrt{t(1 - t)}} \epsilon. \quad (4)$$

The flow-matching objective supervises the network to predict this velocity:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{z_0, \tilde{z}_1, t, \epsilon} \rho(v_{\theta}(z_t, t, s), u_t). \quad (5)$$

Here ρ is the pointwise velocity penalty. The implementation supports MSE, Huber, and L_1 variants in the direct velocity-regression path. Equation (5) should therefore be read as the parent bridge formulation. The current Distinct5 tokenizer family emphasized later in the paper resolves instead to the OMF objective path, where the active transport term is closer to endpoint-delta prediction than to the standalone random-time residual above. This is exactly why the paper keeps its metric claim narrow: the present evidence clearly closes the endpoint-side OT + W_1 terminal-matching story, while the best local transport residual remains an open experiment rather than a settled theorem.

Architecture. The time-conditioned backbone uses sinusoidal time embeddings added to the style code. The implementation is a U-Net-like latent trunk with four semantic cross-attention body blocks at 16×16 resolution. The target style is represented by a learnable spatial prior $P_s \in \mathbb{R}^{128 \times 16 \times 16}$. Semantic cross-attention injects the style prior into content features, while skip connections preserve spatial structure.

Style tokenizer as executable control

The style-side interface is deliberately separated from the latent renderer. In the main domain-level protocol,

the tokenizer receives only a target style identifier and learned style parameters available at inference:

$$c_s = T_\phi(s), \quad v_\theta = F_\theta(z_t, t, c_s). \quad (6)$$

It does not read a target image or target latent from the current evaluation batch. Target-domain latents remain on the loss side of the graph through OT coupling and SA-SWD, but they are not conditioning evidence for T_ϕ . This boundary is important: otherwise training would solve a reference-guided problem while evaluation reports a style-id problem. In this sense, T_ϕ names where transport should go, while F_θ decides how much latent motion is actually needed to get there.

The simplest tokenizer is a per-style code table. Our current implementation generalizes this interface without changing the LANCET consumer:

$$c_s = p_s + \gamma \sum_{k=1}^K \alpha_{s,k} a_k, \quad \alpha_s = \text{softmax}(\ell_s / \tau), \quad (7)$$

where p_s is an optional direct prototype, $\{a_k\}$ are shared style atoms, and γ controls the residual atom contribution. Other tested variants replace p_s with class-local prototypes or use a purely shared VQ-style codebook. The tokenizer therefore exposes a measurable style vocabulary while preserving the single-code interface needed by the existing renderer.

For execution, the backbone may also use style spatial priors and content-conditioned routing. These modules modulate where a target-style control is applied, not what target style is requested. This distinction matches the empirical diagnosis: increasing tokenizer capacity alone did not break the style ceiling, while prototype-aware target queues and content-guided routing improved the executed style-content frontier. We therefore treat tokenizer design as an executable representation problem: the code geometry must remain separable, and the generated latent residuals must also become separable after the frozen or jointly trained LANCET consumes the code.

Complementary roles. The components address distinct subproblems. OT coupling provides target-style endpoints for unpaired supervision, but not transport dynamics. Transport supervision learns how to move toward those endpoints, but by itself does not guarantee that the integrated endpoint matches target-domain texture statistics. SA-SWD regularizes this endpoint mismatch, while semantic cross-attention provides local routing for where style is written and which target patch axes drive the terminal distribution match. The destructive ablations directly verify the roles of terminal distribution matching and kinetic regularization; the planned Random-SWD versus SA-SWD ablation isolates whether semantic projection-axis selection is necessary beyond ordinary endpoint SWD.

Semantic-aligned terminal distribution matching

In addition to flow-matching supervision, a terminal loss regularizes the Euler-integrated endpoint toward

the target style distribution. Let $\Phi_\theta(z_0, s)$ denote the endpoint obtained by integrating v_θ from z_0 with style s . The terminal loss is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1(\Phi_\theta(z_0, s), \tilde{z}_1), \quad (8)$$

where the SWD compares generated endpoint patches with coupled target patches. In the main configuration, these projections are semantic-aligned rather than purely random. Let K_s denote the keys produced by semantic cross-attention for target style s . We rank spatial target positions by the average magnitude of K_s and use the corresponding target latent patch vectors as normalized projection directions. This yields SA-SWD: terminal distribution matching is still computed by sorted one-dimensional projections, but the projection axes are selected from the same semantic routing signal that injects style into the backbone.

The complete training objective combines transport supervision, terminal distribution matching, and kinetic path regularization:

$$\mathcal{L} = \mathcal{L}_{\text{transport}} + \lambda_{\text{term}} \mathcal{L}_{\text{term}} + \lambda_{\text{kin}} \mathcal{L}_{\text{kin}}, \quad (9)$$

where $\mathcal{L}_{\text{transport}}$ is instantiated either as the bridge-formulation penalty in Eq. (5) or as the current OMF-style endpoint-delta transport term, depending on the resolved objective family. where $\mathcal{L}_{\text{kin}} = \mathbb{E} \|v_\theta\|_2^2$ penalizes excessive latent displacement. The primary configuration uses $\lambda_{\text{term}} = 20.0$ and $\lambda_{\text{kin}} = 1.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This three-part loss separates the forces that drive style from those that preserve content.

Metric choice in latent space. The training objective does not treat all latent discrepancies as Euclidean reconstruction errors. The kinetic term uses an L_2 penalty on velocity because it represents path energy, while terminal endpoint matching uses Wasserstein-1 style discrepancies over sorted projections. This separation is important in compressed VAE latents: point-wise endpoint MSE would reward mean solutions and over-penalize semantically valid local rearrangements, whereas SA-SWD preserves distributional supervision without requiring paired spatial correspondence. The current reproduced Distinct5 checkpoints do not enforce style alignment through latent endpoint MSE. Their style pressure is instead carried by OT endpoint construction plus W_1 -style terminal matching, which is the more consequential correction in our measured setting. We therefore treat the unresolved choice of local transport penalty as an experiment-design question, not as a closed headline claim.

Theoretical grounding of latent transport. The transport design admits a lightweight design-grounding analysis (Sec.): Theorem 1 connects the training kinetic loss to the inference-time path action, Theorem 2 bounds the Euler discretization error at $O(1/K)$, and Theorem 3 provides a Lipschitz-continuity rationale for OT-based endpoint coupling. Each result is paired with direct empirical validation.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^M$, the empirical terminal objective is

$$\mathcal{L}_{\text{swd}} = \sum_{r \in \mathcal{R}} \frac{1}{M} \sum_{m=1}^M \left\| \text{sort}(\mathcal{P}_r(\hat{z}_1)\theta_m) - \text{sort}(\mathcal{P}_r(z_s)\theta_m) \right\|_1. \quad (10)$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics along semantic routing directions, not whether it matches any particular artwork spatially.

Formal analysis

We provide three formal results that characterize key properties of the proposed method. Full proofs appear in the supplement. For brevity we state the results and their empirical support.

Notation. Recall the OMF training regime: $\hat{v} = v_\theta(z_0, 1, s)$, $\hat{z}_1 = z_0 + \hat{v}$. At inference, integration follows $dz/dt = v_\theta(z, t, s)$ with K -step Euler: $z_{k+1} = z_k + \Delta t v_\theta(z_k, t_k, s)$ where $\Delta t = 1/K$, $t_k = (k + 0.5)\Delta t$. Define the path action $\mathcal{A}(v) = \int_0^1 \mathbb{E} \|v_\theta(z(t), t, s)\|^2 dt$ and the training kinetic loss $\mathcal{L}_{\text{kin}} = \mathbb{E} \|v_\theta(z_0, 1, s)\|^2$.

Scope and limitation. Theorem 1 should be read as a local design justification, not as a global characterization of arbitrary learned trajectories. The result assumes bounded temporal drift and Lipschitz continuity, and therefore explains when endpoint kinetic regularization is a useful proxy for path energy. It does not claim that a single endpoint penalty exactly controls the full trajectory in all regimes. This is why we pair the theorem with empirical path-statistics measurements rather than presenting it as a closed-form solution of the global latent transport problem.

Theorem 1 (Path action decomposition). Assume that for a fixed content z_0 the velocity norm varies at most δ with t , i.e., $\sup_t \left| \|v_\theta(z_0, t, s)\| - \|v_\theta(z_0, 1, s)\| \right| \leq \delta$, and that v_θ is L -Lipschitz in z . Let M bound $\|v_\theta\|$ along the trajectory. Then the path action decomposes as

$$\mathcal{A}(v) = \mathcal{L}_{\text{kin}} + \varepsilon_t + \varepsilon_z, \quad (11)$$

where $|\varepsilon_t| \leq 2\delta\sqrt{\mathcal{L}_{\text{kin}}} + \delta^2$ and $|\varepsilon_z| \leq LM$. If δ and LM are small, the inference-time path energy is approximated by the training kinetic loss.

Empirical support. Experimentally (128-step integration, D0 model): $\mathcal{L}_{\text{kin}} \approx 1017$, $\mathcal{A}(v) \approx 698$, yielding $|\varepsilon_t + \varepsilon_z| \approx 319$. Removing $\mathcal{L}_{\text{kin}}$ (D2) reduces the path length ratio from 0.98 to 0.94, confirming the kinetic term controls path regularity.

Theorem 2 (Euler discretization bound). Under the L -Lipschitz condition on v_θ , the K -step Euler integration error satisfies

$$\|z(1) - z_K\| \leq \frac{C}{K}, \quad C = \frac{1}{2}(ML_t + M^2L) \frac{e^L - 1}{L}, \quad (12)$$

where L_t bounds the t -derivative of v_θ .

Empirical support. Measured over 160 content latents (D0 model, 256-step reference): $\|z_K - z_{256}\| = \{10.7, 3.80, 1.74, 0.84, 0.40, 0.19, 0.084, 0.027\}$ for $K = 1, 2, 4, 8, 16, 32, 64, 128$; the error halves when K doubles, confirming the $O(1/K)$ scaling. At $K = 12$ (the default), the error is < 0.6 —negligible relative to the style-transfer effect.

Theorem 3 (OT assignment continuity). Let the SWD cost $c(\cdot, \cdot)$ be L_c -Lipschitz and let π^* be the ε -Sinkhorn plan on a batch of size B . Let D denote the diameter of the target latent set. Then, for any two content latents z_0, z'_0 , the expected matched targets satisfy

$$\|\mathbb{E}_{\pi^*}[\tilde{z}_1 | z_0] - \mathbb{E}_{\pi^*}[\tilde{z}_1 | z'_0]\| \leq \frac{D \cdot L_c}{\varepsilon} \|z_0 - z'_0\| + O(\varepsilon). \quad (13)$$

Thus the OT assignment map is (DL_c/ε) -Lipschitz: nearby content latents receive nearby matched targets. Random uniform coupling provides no such guarantee. The Lipschitz property implies smoother supervision gradients and more directionally coherent displacement vectors across the batch.

Empirical support. Compared over 640 content latents: OT coupling yields mean cosine similarity to the batch-mean direction of 0.150, vs. 0.124 for random coupling, a 21% improvement. The transport cost is also reduced by 4.2% (0.119 vs. 0.124).

Experiments

Protocol hierarchy and metrics

We organize the evidence into two non-interchangeable result families. This separation is necessary because the same scalar pair, CLIP-style and LPIPS, can be misleading when datasets, image resolution, or baseline training protocols change.

Historical strict-750. The main legacy protocol uses five domains—photo, Hayao, Monet, Van Gogh, and Cezanne—and evaluates all 5×5 source-target directions with 30 source images per source domain, for 750 generated outputs. This is the protocol used for comparisons to SaMST, S2WAT, StyleID, AdaIN, StyTr2, CAST, AesFA, and AesPA-Net.

Distinct5-512 stress split. The new stress split uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These five classes were chosen as the most strongly separated subset under a CLIP-style screening pass over WikiArt categories, rather than being hand-picked for LBM. Evaluation again uses all 5×5 directions over 750 generated

images. This split complements the historical table with a stronger domain-separation test and an explicit identical-image baseline.

Internal convergence reference. We also track a broader WikiArt512 five-style setup during development, but we do not use it for headline claims because the identical-image CLIP-style prior is too high and the resulting conclusions are qualitatively similar to the historical 256/strict-750 convergence story.

We report CLIP-style (CLIP-S) and CLIP-content (CLIP-C) using CLIP ViT-B/32 embeddings (Radford et al. 2021), together with LPIPS-content and

$$EC = \text{CLIP-style} \cdot (1 - \text{LPIPS}). \quad (14)$$

For Distinct5-512 we additionally report no-op-adjusted style gain,

$$\Delta_{\text{idt}} = \text{CLIP-S}(\text{method}) - \text{CLIP-S}(\text{idt}), \quad (15)$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. We use this quantity to define the metric-stress regime exposed by Distinct5-512: a method is suspicious under raw CLIP-style if it either fails to exceed the identical-image prior ($\Delta_{\text{idt}} \leq 0$) or exceeds it only by moving into a clearly damaged LPIPS region while also deteriorating broad artifact diagnostics. In this paper, CLIP-S is interpreted as target-style affinity, LPIPS as content displacement, and ArtFID as a broader art-domain realism and artifact diagnostic rather than a direct style-gain score. EC is used only as a Pareto summary; claims are checked against CLIP-S and LPIPS individually. For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, shallow Gram micro statistics, and KID where available. All strict-750 baselines use the same source images, target domains, preprocessing, and identity directions. For arbitrary-style methods, one target-domain reference image is supplied per target style; for style-id LBM, the model receives only the target domain identifier. The ‘idt’ control is introduced as a diagnostic baseline for art-to-art transfer rather than as a community standard: our claim is that Distinct5 makes explicit the role of the unchanged-image prior on this split, not that any single existing metric should be discarded.

Historical strict-750 comparison

Table 1 establishes LBM as the strongest compact content-preserving operating point among the reproduced baselines. StyleID reaches the highest raw CLIP-style but collapses content. SaMST is marginally higher on CLIP-S and CLIP-C, yet LBM improves LPIPS and EC while also winning the artifact-sensitive metrics reported below. Relative to S2WAT and AdaIN, LBM preserves substantially more content at comparable or stronger style. The main result is therefore a frontier claim: LBM delivers the cleanest balance of style strength, content fidelity, artifact control, and retraining cost among the compact retrainable systems in this comparison.

Cost matters in two different ways here. First, LBM reaches its selected operating point earlier than the strongest high-quality baselines in the preserved timing records available to us: 310s for the reported LBM checkpoint versus 6769s for the reproduced SaMST operating point and 10600s for S2WAT. We report these as operating-point wall-clock observations, not as a universal time-to-quality theorem, because stopping rules differ across methods. Second, its inference cost remains practical rather than extreme. LBM is not the fastest feed-forward path in absolute milliseconds per image, since lightweight statistic-alignment models such as AdaIN are faster, but it is much faster than StyleID and avoids the multi-hour training cost of SaMST and S2WAT. The practical conclusion is therefore narrower: the proposed transport objective is competitive in quality while remaining inexpensive to retrain and straightforward to evaluate at scale under the present protocol.

Artifact-sensitive diagnosis

The qualitative failure modes explain why the main table is not reduced to a single scalar. SaMST is the closest efficient baseline under the core metrics, but matched grids and texture crops show muddy, grain-like surface artifacts. These artifacts can coexist with good CLIP-content and LPIPS because low-frequency object boundaries remain visible.

Table 3 shows that LBM is not merely winning by leaving the image unchanged. Relative to SaMST, it improves MUSIQ from 36.10 to 49.21, MANIQA from 0.314 to 0.406, DISTS-content from 0.294 to 0.248, HF-Patch-KID from 6.76 to 4.17, and FFT slope error from 1.054 to 0.547. A paired bootstrap over the 750 per-image scores for Ours gives 95% intervals of $[0.4467, 0.4563]$ for LPIPS and $[0.7091, 0.7234]$ for CLIP-S. The photo-to-art subset ($n = 120$) yields CLIP-S 0.6716 and LPIPS 0.4882; the non-identity subset ($n = 600$) yields CLIP-S 0.6911 and LPIPS 0.4609.

Distinct5-512 stress benchmark

Distinct5-512 is the primary 512-based stress benchmark in this paper. Figure 4 is the main result view, and Table 4 gives the exact numbers. The comparison is stricter than a single target direction: every method must handle identity directions, nearby art-to-art transfers, and distant cross-style transfers. The split was chosen using CLIP-style separation and is therefore treated only as a stress test for metric behavior, not as a universal art benchmark. On this split, the idt baseline already reaches CLIP-S 0.6801 while doing no transfer at all, so raw CLIP-style alone overstates performance. Against that explicit floor, LBM occupies the strongest measured content-preserving frontier among the evaluated methods. The content-preserving LBM-F checkpoint reaches CLIP-S 0.6969 and LPIPS 0.3186 after about 1.2 minutes, compared with SaMAM-2250 (Liu et al. 2025) at CLIP-S 0.5811 and LPIPS 0.3538 after about 7.6 hours. This is an operating-point observation rather than a universal claim about SaMAM:

Table 1: Historical strict-750 comparison in grouped family format. Higher is better for CLIP-S, CLIP-C, EC, MUSIQ, and MANIQA; lower is better for LPIPS, DISTS, HF-KID, and KID.

Metric	CNN-based			Transformer-based			Training-free		Latent transport
	AesPA	AesFA	AdaIN	StyTr2	S2WAT	SaMST	CAST	StyleID	LBM
CLIP-S $\uparrow$	0.5976	0.6496	0.7130	0.6374	0.7139	0.7194	0.6652	0.7597	0.7161
CLIP-C $\uparrow$	0.5009	0.5497	0.6990	0.6001	0.7465	0.8193	0.5562	0.5519	0.8086
LPIPS $\downarrow$	0.8215	0.7392	0.6298	0.6345	0.5263	0.4664	0.7263	0.7497	0.4514
EC $\uparrow$	0.1067	0.1694	0.2639	0.2330	0.3382	0.3839	0.1821	0.1902	0.3928
MUSIQ $\uparrow$	—	—	—	—	36.5256	36.0950	—	—	49.2059
MANIQA $\uparrow$	—	—	—	—	0.1754	0.3139	—	—	0.4057
DISTS $\downarrow$	—	—	—	—	0.2942	0.2943	—	—	0.2477
HF-KID $\downarrow$	—	—	—	—	12.6623	6.7598	—	—	4.1694
KID $\downarrow$	—	—	—	—	0.0567	0.0489	—	—	0.0524

Table 2: Training and inference cost on the historical strict-750 protocol. All timings are wall-clock on a single RTX 4070 Laptop GPU. Training time is reported to the selected operating point used in this paper; inference reports 750-image end-to-end runtime and milliseconds per image.

Method	Train (s)	Infer-750 (s)	ms/img	Params
LBM	310	85.4	114	3.91M
SaMST	6769	39.8	53	~6M
S2WAT	10600	—	—	~7M
CAST	1760	75.5	101	7.01M
StyleID	0	603.3	804	—
AdaIN v32k	9220	9.3	12	—

on separated art-to-art transfer, absolute CLIP-S is hard to interpret without an unchanged-image reference. SaMST-512 shows a different failure mode: raw style rises to 0.7247, but LPIPS degrades to 0.6255 and ArtFID worsens to 395.7. The style-seeking LBM-K checkpoint raises CLIP-S to 0.7010 while remaining far below that high-damage region. In no-op-adjusted terms, LBM remains positive under both scopes: LBM-F gives $\Delta_{\text{idt}} = +0.0168$ full and $+0.0244$ transfer-only, while LBM-K gives $+0.0209$ full and $+0.0312$ transfer-only. By contrast, SaMAM stays below the idt floor throughout the evaluated run, and SaMST reaches positive style gain only in a high-LPIPS / high-ArtFID regime on this split.

Tokenizer and representation ablations

These ablations make the tokenizer conclusion sharper than a parameter-count story. Class-local prototypes and global atoms did not beat the baseline frontier, so merely enlarging the tokenizer is insufficient. Content-guided spatial routing helped, which indicates that execution location matters. The largest improvement came from a prototype-aware latent target queue, which reduces target-distribution noise before the terminal loss

Table 3: Artifact-sensitive comparison among high-quality methods.

Metric	Ours e7	SaMST	S2WAT
MUSIQ $\uparrow$	49.2059	36.0950	36.5256
MANIQA $\uparrow$	0.4057	0.3139	0.1754
DISTS-content $\downarrow$	0.2477	0.2943	0.2942
HF-Patch-KID $\downarrow$	4.1694	6.7598	12.6623
FFT slope error $\downarrow$	0.5473	1.0536	0.7017
Gram micro $\downarrow$	0.0798	0.0947	—

Gram micro was not recomputed for S2WAT in the current evaluation protocol.

Table 4: Distinct5-512 strict-750 stress benchmark. All rows use the same 5×5 all-pairs evaluation over 750 generated images. ‘idt’ is the unchanged-image reference copied into every target slot. Δ_{idt} is the full-scope style gain over idt and $\Delta_{\text{idt},\text{tr}}$ is the transfer-only off-diagonal gain. SaMAM and SaMST training time is cumulative wall time with evaluation excluded; LBM times are per-checkpoint remote RTX 3060 training wall times.

Method	CLIP-S $\uparrow$	LPIPS $\downarrow$	EC $\uparrow$	ArtFID $\downarrow$	$\Delta_{\text{idt}} \uparrow$	$\Delta_{\text{idt},\text{tr}} \uparrow$	Train
idt (no-op)	0.6801	0.0000	0.6801	216.5	0.0000	0.0000	0
SaMAM 2000	0.5833	0.3622	0.3721	152.2	-0.0968	-0.0853	6.8h
SaMAM 2250	0.5811	0.3538	0.3755	146.1	-0.0990	-0.0877	7.6h
SaMST-512 e15	0.7247	0.6255	0.2714	395.7	+0.0446	+0.0558	5.8h
LBM-F e1	0.6969	0.3186	0.4748	122.6	+0.0168	+0.0244	1.2m
LBM-K e1	0.7010	0.3623	0.4470	157.2	+0.0209	+0.0312	1.2m

is applied. Finally, content-adaptive atom routing produced the best style score but regressed LPIPS, suggesting that the representation can increase style separability only by asking the vector field to move farther. The next tokenizer should therefore expose two factors rather than one larger code: a target-style carrier that remains separable after execution, and a content-risk gate that bounds how much of that carrier should be applied.

Non-training probes support the same diagnosis. On the historical e8 checkpoint, tokenizer codes have near-

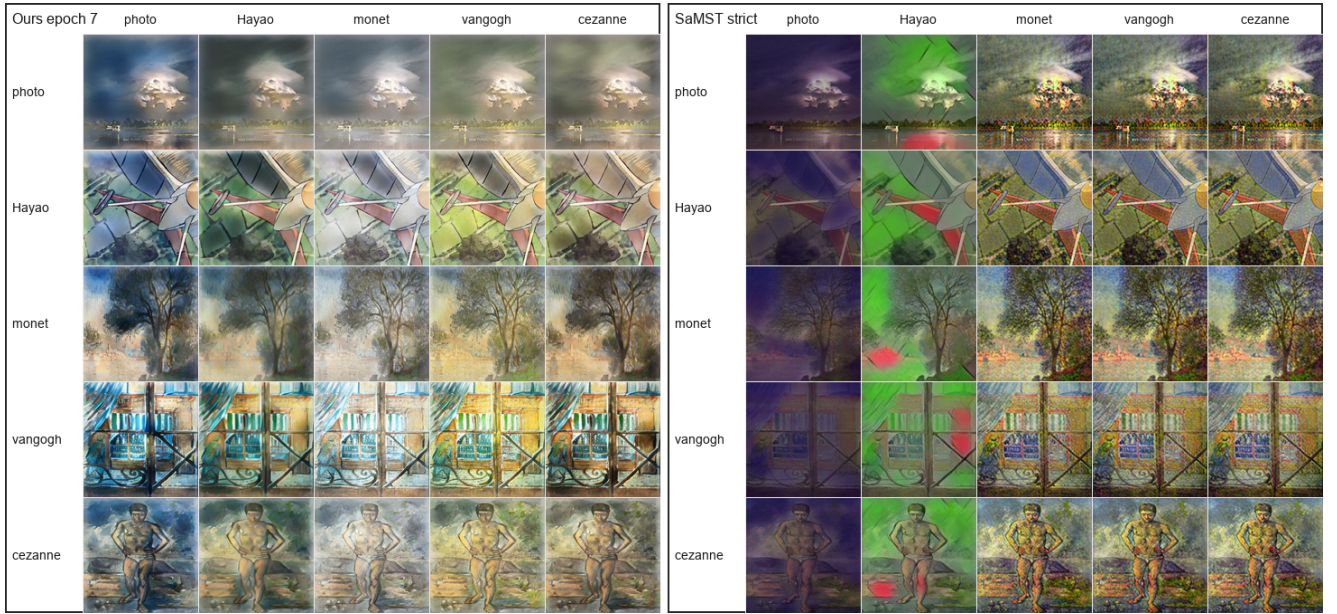


Figure 2: Qualitative 5-by-5 source-target grids for Ours epoch 7 and SaMST strict. Rows indicate source domains and columns indicate target domains. SaMST preserves coarse structure but often produces muddy, grain-like surface texture; Ours is less aggressive in raw style but visually cleaner.

Table 5: Distinct5-512 tokenizer and representation ablations. The results show that capacity alone is not the issue; target selection, content-guided execution, and sparse routing determine whether the representation is executable.

Variant	Best epoch	CLIP-S	LPIPS	Interpretation
Direct atom residual baseline	8/1	0.6876	0.4468	weak baseline; simple residual atoms do not solve representation
Class-local prototypes	8/1	0.6849	0.4464	more per-class capacity, no better style geometry
Global VQ atoms	8	0.6873	0.4446	small LPIPS gain, no style breakthrough
Content-guided spatial routing	2	0.6907	0.4226	first simultaneous style and LPIPS gain
VQ + content-guided routing	1	0.6898	0.4156	stronger content preservation, style still limited
Prototype-aware latent target queue	1/3	0.6973	0.3331	major jump from better internal target distribution
Annealed prototype queue (F)	1	0.6969	0.3186	current LPIPS reference point
Hard-explore prototype queue (H)	2/1	0.6994	0.3213	current balanced point
Content-adaptive VQ atom routing (K)	1	0.7010	0.3623	style boost, but larger endpoint movement

full rank for five styles (effective rank 3.986, mean off-diagonal cosine 0.015), while generated latent residuals are much more aligned (effective rank 3.324, mean off-diagonal cosine 0.725). Residual variance is also content dominated: source-image variance explains 0.9501 of raw generated-delta variance, while target style explains only 0.0283 before source-image removal but 0.5664 after removing per-source-image residual means. Thus the immediate bottleneck is not raw tokenizer collapse; it is making target-style control survive the content-conditioned renderer.

Mechanism ablations

Removing terminal distribution matching gives excellent content preservation but weak style, showing that endpoint distribution alignment is the primary style driver. Removing kinetic regularization preserves style but collapses content, confirming that path-energy regularization is essential. Strong color loss damages the

Table 6: Destructive ablations under the historical strict-750 protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o SA-SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

trade-off, so naive color matching is not used in the mainline. Trajectory straightness analysis gives the same reading: the full model has PLR 0.98, while removing SA-SWD produces near-linear transport to the OT target (PLR 0.9995) and removing kinetic regularization reduces straightness to PLR 0.94 with content collapse.

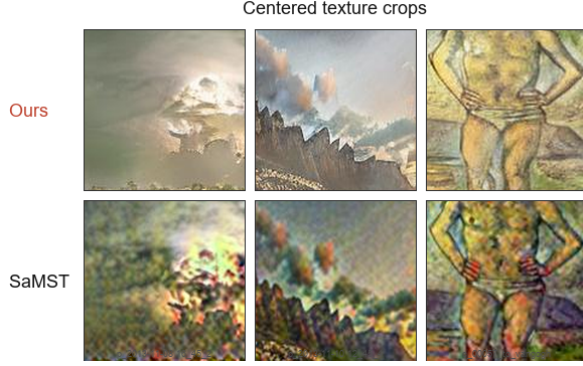


Figure 3: Centered texture crops from matched Ours and SaMST outputs, isolating the local surface-texture failure mode.

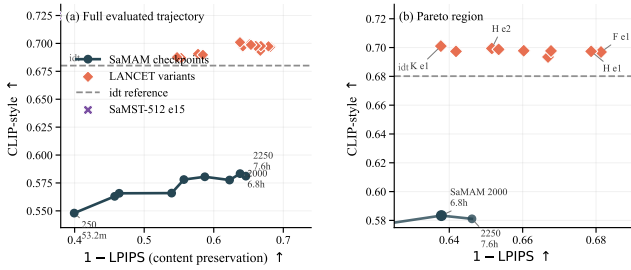


Figure 4: Distinct5-512 style-content frontier. The dashed horizontal line marks the idt no-op CLIP-style level. LBM points are the best tokenizer-queue variants, the SaMAM curve shows cumulative training progress, and the SaMST-512 point marks a high-style, high-damage regime.

Sensitivity and efficiency

The 40-run category-weight sweep trained 20 sampling recipes with $K \in \{1, 2\}$ for eight epochs each, evaluating every epoch for 320 total checkpoints. The best EC result was K2 balanced default at epoch 3 with CLIP-S 0.6980, CLIP-C 0.8727, LPIPS 0.3777, and EC 0.4343. The best raw-style result in the sweep was K1 balanced default at epoch 8 with CLIP-S 0.7161, CLIP-C 0.7984, LPIPS 0.4605, and EC 0.3863. The sweep showed that more complex category weighting was not consistently better than balanced sampling.

Step-count sensitivity is also stable. Evaluating 1, 4, 8, 12, and 16 Euler steps on the selected epoch-7 checkpoint changes all-pairs CLIP-S only from 0.7159 to 0.7162, while LPIPS remains near 0.4514. Four steps already recover nearly identical quality, with measured strict-750 throughput increasing from 8.43 images/s at one step to 9.54 images/s at four steps and remaining around 9.34–9.50 images/s for 8–16 steps. A 3×3 grid over $\lambda_{\text{kin}} \in \{0.5, 1.0, 2.0\}$ and $\lambda_{\text{swd}} \in \{10, 20, 30\}$ produced a smooth style-content surface: higher kinetic weight and lighter terminal pressure improve content, while weaker kinetic weight or stronger terminal pres-

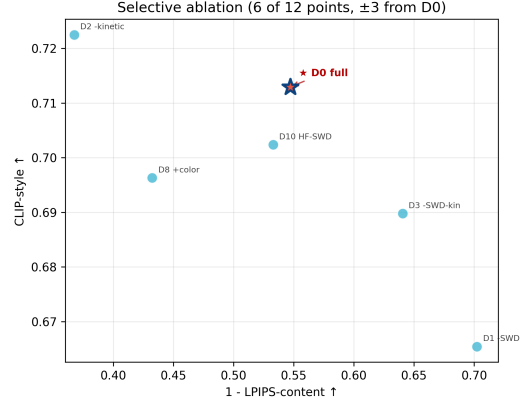


Figure 5: Destructive ablations. Removing SA-SWD improves content but weakens style, while removing kinetic regularization increases style at the cost of severe content degradation.

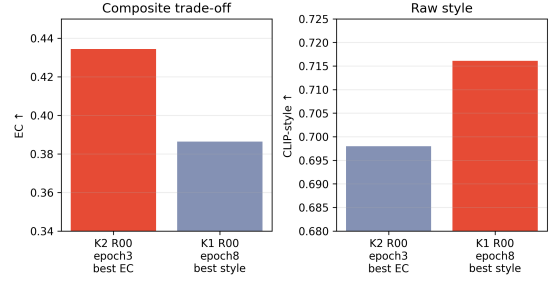


Figure 6: Summary of the 40-run category-weight sweep. K2 balanced default gives the best EC, whereas K1 balanced default gives the best raw CLIP-style among evaluated checkpoints.

sure increases style at the cost of LPIPS.

All reported evaluations use fixed seeds, stored resolved configurations, source snapshots, checkpoints, and per-epoch logs. Historical timings were measured on an RTX 4070 Laptop GPU unless otherwise noted. Distinct5-512 LBM and SaMAM timings are reported separately from the remote RTX 3060 WSL setup, with SaMAM time cumulative over training and LBM time per checkpoint. For the historical table, training and inference cost are reported explicitly in Table 2 rather than being folded into a single efficiency claim.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. On the historical strict-750 protocol, StyleID has the highest raw style but collapses content, SaMST is slightly stronger than LBM on CLIP-S and CLIP-C but shows grain-like local artifacts, and LBM occupies a cleaner LPIPS/artifact-sensitive operating point. Distinct5-512 sharpens the same lesson further because its no-op baseline is measured explicitly: higher raw style is achievable, but

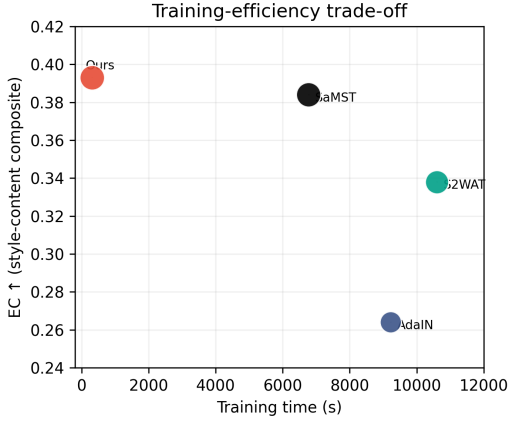


Figure 7: Training-efficiency Pareto plot (log scale). Ours operates in a low-training-cost region while preserving a competitive style-content frontier. Marker size reflects parameter count.

only by moving into a severe LPIPS and ArtFID failure region, and some baselines fail even to exceed the identical-image prior. This is best read as a bounded evaluation diagnosis, not as a blanket rejection of prior baselines. Earlier AST evaluation work already notes that protocols are inconsistent across papers and that automatic scores can be style-dependent or misaligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our narrower claim is that art-to-art transfer benefits from reporting an unchanged-image reference that measures how much target-style similarity is already present before stylization, especially on separated art-to-art splits. Across both protocols, the consistent conclusion is a frontier result: LBM gives the strongest measured content-preserving operating region, while the remaining weakness is faithful style execution rather than style separability in the tokenizer alone.

The tokenizer experiments narrow the design question. A larger style code or more class-local prototypes is not sufficient. The measured bottleneck is that target-style information must survive execution through a content-conditioned vector field. The current evidence favors target-style carriers, prototype-aware target queues, and content-aware routing over black-box token capacity. It also shows a trade-off: content-adaptive atom routing can increase style strength but may increase endpoint movement and LPIPS. A mature next-stage tokenizer should therefore factor target representation from execution budget rather than simply adding parameters.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher reso-

lutions may need better solvers or stronger constraints. Third, SA-SWD is an efficient empirical distribution loss, not an exact global transport solver. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. Finally, Distinct5-512 is a five-domain stress split rather than a universal art benchmark, so broader domain coverage remains future work even though the current cross-method comparison is already decisive on this split.

Conclusion

We presented Latent Bridge Matching, an OT-coupled latent flow framework for efficient domain-level artistic transfer. The method separates endpoint construction, path-wise flow learning, semantic terminal distribution matching, and style-tokenized execution. The historical strict-750 results show that LBM reaches a cleaner style-content trade-off than SaMST, S2WAT, StyleID, and AdaIN-family baselines while remaining inexpensive to train and practical to evaluate. The Distinct5-512 stress benchmark sharpens the central insight: once the identical-image prior is made explicit, LBM stays above the no-op floor and defines the strongest measured content-preserving frontier on a harder separated-domain split. The tokenizer study then narrows the next design target. The bottleneck is not raw token capacity, but executing a separable style carrier through a content-conditioned renderer without paying an LPIPS penalty. The next representation stage should therefore factor target-style content from execution budget through an explicit target carrier plus content-risk gate.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: Yes.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: Yes.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: Yes.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides three formal results with proofs and experimental validation (Sec.): a path-action decomposition connecting the training kinetic loss to inference-time path energy (Theorem 1), an $O(1/K)$ Euler discretization error bound (Theorem 2), and a Lipschitz-continuity result for the OT assignment map explaining its directional-coherence benefit (Theorem 3). Full proofs are in the supplement. These results justify the three-part design without claiming equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: Yes.
- All novel datasets introduced in this paper are included in a data appendix: NA.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: NA.
- All datasets drawn from the existing literature are accompanied by appropriate citations: Yes.
- All datasets drawn from the existing literature are publicly available: Yes.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying: NA.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: Partial.
- Any code required for pre-processing data is included in the appendix: No.
- All source code required for conducting and analyzing the experiments is included in a code appendix: No.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: Partial.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: Partial.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: Partial.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: Partial.

- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: Yes.
- This paper states the number of algorithm runs used to compute each reported result: Yes.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: Yes.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: Yes (paired bootstrap, 5000 resamples, reported in Sec. Experiments).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: Partial.